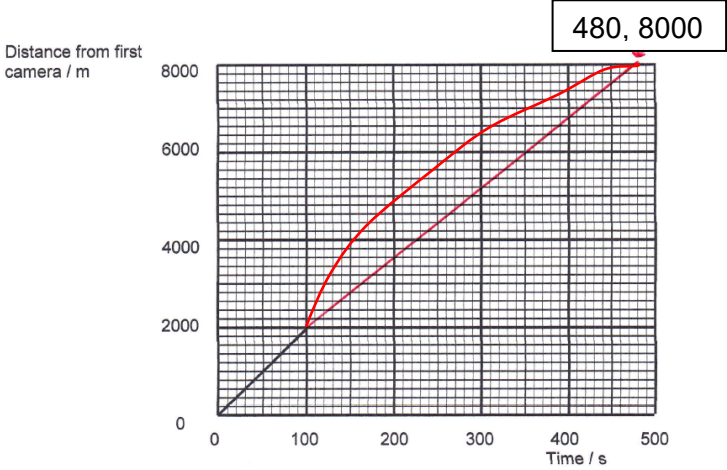


Question			Marking details	Marks available																	
				AO1	AO2	AO3	Total	Maths	Prac												
2	(a)		Vector: Magnitude (size) and direction Scalar: Magnitude (size) only Minimum acceptable response: <i>a vector has direction</i> [1] Relevant example of each e.g. independent mark [1] <table><tr><th>Vectors</th><th>Scalars</th></tr><tr><td>Displacement</td><td>Speed</td></tr><tr><td>Velocity</td><td>Time</td></tr><tr><td>Acceleration</td><td>Distance</td></tr><tr><td>Force</td><td>Pressure</td></tr><tr><td></td><td>Temperature</td></tr></table>	Vectors	Scalars	Displacement	Speed	Velocity	Time	Acceleration	Distance	Force	Pressure		Temperature	2			2		
Vectors	Scalars																				
Displacement	Speed																				
Velocity	Time																				
Acceleration	Distance																				
Force	Pressure																				
	Temperature																				
	(b)	(i)	From graph speed = $\frac{2\,000}{100}$ or equiv [e.g. $\frac{1000}{50}$] (or 20 ms ⁻¹) seen [1] Convincing conversion to km h ⁻¹ (i.e. $\times \frac{3\,600}{1000}$) or 72 [km h ⁻¹] seen or convert 70 km h ⁻¹ into 19.4 m s ⁻¹ [1]	1	1		2	2													

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(ii)	Time to complete distance at mean speed of $60 \text{ km h}^{-1} = \frac{8}{60}$ [1] (= 0.1333) Conversion $t = 480 \text{ s}$ [1] Accept rounded values e.g. 17 m s^{-1} gives $t = 471 \text{ s}$ Alternative for first 2 marks: $60 \text{ km h}^{-1} = \frac{1000}{60} \text{ m s}^{-1}$ (1) $t = \frac{8000}{\left(\frac{1000}{60}\right)} = 480 \text{ s}$ (1) Alternative for first 2 marks: 1 km per minute (1) $t = 8 \text{ minutes or } 480 \text{ s}$ (1) 3rd mark - Continuous (though not necessarily straight) line drawn on graph from (100, 2000) to (480, 8000) tolerance $\pm$ small square - two possible examples shown: [1] Distance from first camera / m  Time / s		3		3	3	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(c)		Correct substitution into $v^2 = u^2 + 2ax$ (i.e. $0 = u^2 - (2 \times 8 \times 85)$ [1] don't award mark if incorrect signs $u = 36.9 \text{ [ms}^{-1}\text{]}$ ecf on incorrect signs [1] No, greater than speed limit [1] must correlate with working Alternative: Correct substitution into $v^2 = u^2 + 2ax$ using 30 ms^{-1} to find x . (i.e. $0 = (30)^2 - (2 \times 8 \times x)$ [1] don't award mark if incorrect signs $x = 56.3 \text{ [m]}$ ecf on incorrect signs [1] No, greater than speed limit [1] must correlate with working Alternative: Using $v = u + at$ to determine $t = 3.75 \text{ s}$ (1) Using $x = \frac{1}{2}(u + v)t$ i.e. $3.75 \times \text{mean velocity (15)} = 56.3 \text{ [m]}$ (1) No, greater than speed limit [1] must correlate with working			3	3	2	
			Question 2 total	3	4	3	10	7	0